



Sanjivani Rural Education Society's
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Subject- Object Oriented Programming (CO212)
Unit 1 – Fundamentals of OOP
Topic – 1.6 Control Structures in C++

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Control Structures in C++

- **Control structures** are used to affect how statements are executed.

Three Control Structures Are:-

- Sequence structure (Straight line)
- Selection structure (Branching)
- Loop structure (Iteration)



Sequence Structure

- Statements are executed one after the other.

- This is C++

Sequence Example:-

```
#include <iostream>
using namespace std;

int main(){//Global main function.
    //This main function executes a sequence of
    // statements and then terminates. The
    // statements in the sequence do not
    // include decisions or loops.
    cout << "Hello Viet Nam\n";
    cout << "Hello Iraq\n";
    cout << "Hello America\n";
    cout << "Hello World\n";
    return 0;
} //end main function
```



Selection Structure

- It is used to choose among alternative actions.
- Making decisions

Three selection structures are:-

- If
- if/else
- switch



If selection structure

- Used to do something if some condition is “true”.
if syntax:-

```
if (<expression>
    <statement>;
```

Example:

```
if (grade > 60)
    cout << "Passed" << endl;
```



If/else selection structure

- Used to do one thing if some condition is “true” and something else if the condition is “false”.

if/else syntax:-

```
if (<expression>)  
    <statement>;  
else  
    <statement>;
```

Example:

```
if (grade >= 60)  
    cout << "Passed" << endl;  
else  
    cout << "Failed" << endl;
```



Switch Selection structure

- This is a multiple-Branching Statement where based on a condition the control is transferred to one of the many possible points.

Switch case Example:-

```
switch(o) {  
    case '+':  
        cout<<num1<<" + "<<num2<<" = "<<num1+num2;  
        break;  
    case '-':  
        cout<<num1<<" - "<<num2<<" = "<<num1-num2;  
        break;  
    case '*':  
        cout<<num1<<" * "<<num2<<" = "<<num1*num2;  
        break;  
    case '/':  
        cout<<num1<<" / "<<num2<<" = "<<num1/num2;  
        break;  
    default:  
        cout<<"Operator not matched..";  
}
```



Loop structure

- Used to repeat a set of instructions.

- Looping

There are three selection structures available in C++:-

- Do-While loop
- While loop
- For loop



Do-While Loop structure

- The do-while is an exit-controlled loop. Based on a condition the control is transferred back to a particular point in the program.

Do-While Example:-

```
int main()
{
    int x=0;
    do
    {
        cout<<x<<endl;
        x=x+1;
    }while(x<=5)
    return 0;
}
```

Output:

0
1
2
3
4
5



While Loop structure

- The while loop is an entry-controlled loop

While Example:-

```
int main()
{
    int x=0;
    while(x<=5)
    {
        cout<<x<<endl;;
        x=x+1;
    }
    return 0;
}
```

Output:

0
1
2
3
4
5



For Loop structure

- for is an entry-controlled loop and is used when an action is to be repeated for a number of times.

For Example:-

```
int main()
{
    const int UPPER_LIMIT = 5;
    int num;

    for (num = 1; num <=UPPER_LIMIT; num++)
    {
        cout << num << endl;
    }

    return 0;
}
```

Output:

1
2
3
4
5



Task to be perform using appropriate control structure

- Write a program to check number is prime or not.
- Write a program to check number is amstrong number or not.
- Write a program to find palidrome number
- Write a program to print following output:

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

Thank you..